● Source code

```
#include <stdio.h>
#include <stdlib.h>
#include <string.h>

/* Structure Definition */
struct Address {
    char name[50];
    char street[50];
    char city[50];
    char postcode[20];
    char email[50];
    char country[30];
    char state[30];
    struct Address* next;
    struct Address* prev;
};

/* Function Declarations */
struct Address* enter(struct Address* head);
void display(struct Address* head);
void search(struct Address* head, const char* key);
struct Address* deleteAddress(struct Address* head, const char* key);
void save(struct Address* head);
struct Address* load();

/* Add New Record */
struct Address* enter(struct Address* head) {
    struct Address* newAddress = (struct Address*)malloc(sizeof(struct Address));

    printf("Enter Name: ");
    scanf("%s", newAddress->name);
    printf("Enter Street: ");
    scanf("%s", newAddress->street);
    printf("Enter City: ");
    scanf("%s", newAddress->city);
    printf("Enter Postcode: ");
    scanf("%s", newAddress->postcode);
    printf("Enter Email: ");
    scanf("%s", newAddress->email);
    printf("Enter Country: ");
    scanf("%s", newAddress->country);
    printf("Enter State: ");
```

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    scanf("%s", newAddress->state);

    newAddress->prev = NULL;
    newAddress->next = head;

    if (head != NULL)
        head->prev = newAddress;

    printf("Information added successfully.\n");
    return newAddress;
}

/* Display All Records */
void display(struct Address* head) {
    if (head == NULL) {
        printf("Address book is empty.\n");
        return;
    }

    printf("\n===== Address Book =====\n");
    while (head != NULL) {
        printf("Name: %s\nStreet: %s\nCity: %s\nPostcode: %s\nEmail: %s\nCountry:
%s\nState: %s\n\n",
                head->name, head->street, head->city,
                head->postcode, head->email,
                head->country, head->state);
        head = head->next;
    }
}

/* Search Record */
void search(struct Address* head, const char* key) {
    while (head != NULL) {
        if (strcmp(head->name, key) == 0) {
            printf("Record Found:\n");
            printf("Name: %s\nStreet: %s\nCity: %s\nPostcode: %s\nEmail: %s\nCountry:
%s\nState: %s\n",
                    head->name, head->street, head->city,
                    head->postcode, head->email,
                    head->country, head->state);
            return;
        }
        head = head->next;
    }
```

```
        printf("Record not found.\n");
}

/* Delete Record */
struct Address* deleteAddress(struct Address* head, const char* key) {
        struct Address* current = head;

        while (current != NULL) {
                if (strcmp(current->name, key) == 0) {
                        if (current->prev != NULL)
                                current->prev->next = current->next;
                        else
                                head = current->next;

                        if (current->next != NULL)
                                current->next->prev = current->prev;

                        free(current);
                        printf("Record deleted successfully.\n");
                        return head;
                }
                current = current->next;
        }

        printf("Record not found.\n");
        return head;
}

/* Save to File */
void save(struct Address* head) {
        FILE* file = fopen("address_book.txt", "w");
        if (file == NULL) {
                printf("Error opening file.\n");
                return;
        }

        while (head != NULL) {
                fprintf(file, "%s %s %s %s %s %s %s\n",
                                head->name, head->street, head->city,
                                head->postcode, head->email,
                                head->country, head->state);
                head = head->next;
        }
```

```
    fclose(file);
    printf("Data saved successfully.\n");
}

/* Load from File */
struct Address* load() {
    FILE* file = fopen("address_book.txt", "r");
    if (file == NULL) {
        printf("No saved file found.\n");
        return NULL;
    }

    struct Address* head = NULL;
    struct Address* tail = NULL;

    while (1) {
        struct Address* newAddress = (struct Address*)malloc(sizeof(struct Address));
        int result = fscanf(file, "%s %s %s %s %s %s %s",
                                  newAddress->name, newAddress->street,
                                  newAddress->city, newAddress->postcode,
                                  newAddress->email, newAddress->country,
                                  newAddress->state);

        if (result != 7) {
            free(newAddress);
            break;
        }

        newAddress->next = NULL;
        newAddress->prev = tail;

        if (head == NULL)
            head = newAddress;
        else
            tail->next = newAddress;

        tail = newAddress;
    }

    fclose(file);
    printf("Data loaded successfully.\n");
    return head;
}
```

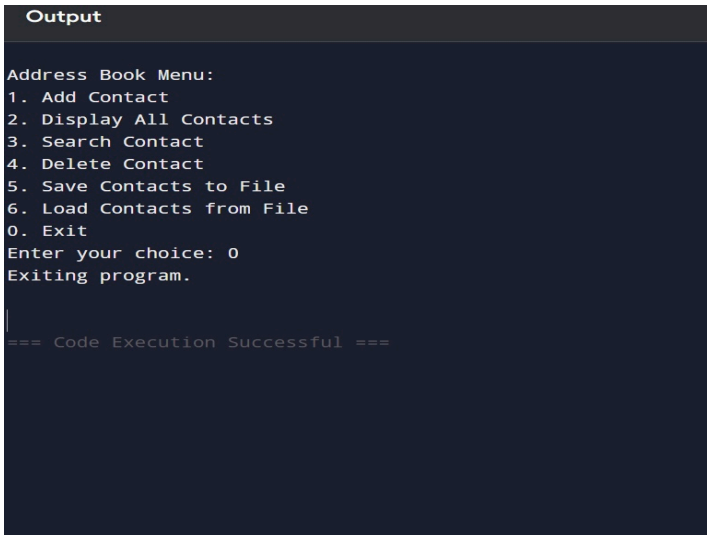
```
/* Main Function */
int main() {
    struct Address* head = NULL;
    int choice;
    char name[50];

    while (1) {
        printf("\n====== Address Book Menu ======\n");
        printf("1. Enter Information\n");
        printf("2. Display Information\n");
        printf("3. Search Information\n");
        printf("4. Delete Information\n");
        printf("5. Save to File\n");
        printf("6. Load from File\n");
        printf("0. Exit\n");
        printf("Choose an option: ");
        scanf("%d", &choice);

        switch (choice) {
        case 1:
            head = enter(head);
            break;
        case 2:
            display(head);
            break;
        case 3:
            printf("Enter name to search: ");
            scanf("%s", name);
            search(head, name);
            break;
        case 4:
            printf("Enter name to delete: ");
            scanf("%s", name);
            head = deleteAddress(head, name);
            break;
        case 5:
            save(head);
            break;
        case 6:
            head = load();
            break;
        case 0:
            printf("Exiting program...\n");
            return 0;
```

```
            default:
                printf("Invalid option.\n");
            }
        }
}
```

- **output**



- **Program Name**: Address Book Management System
- **Description**:

This program is an address book management system implemented in C using a doubly linked list.

It allows the user to add, display, search, delete, save, and load address records.

Each record contains personal and address information such as name, street, city, postcode, email, country, and state.

The program also supports file storage to save and reload data.

Language: C

Data Structure Used: Doubly Linked List

Compiler: GCC